

1. The OSI Reference Model

The foundation of ONGC's network infrastructure operates on the **OSI Model**, breaking communication into 7 layers: [\[1, 2, 3\]](#)

- **Physical & Data Link Layers:** Responsible for hardware (Ethernet, Fiber Optics), Network Interface Cards (NICs), and MAC addresses.
- **Network Layer:** Uses the **TCP/IP** protocol and routes data packets using IP addresses.
- **Transport, Session & Presentation Layers:** Manages port connections, data flow control, encryption, and data formatting.
- **Application Layer:** Provides direct user interface for systems like **SAP, SCADA, and LOTUS**. [\[1, 2, 3, 4, 5\]](#)

2. Core Network Devices

All data travels through these essential devices: [\[1\]](#)

- **Routers:** Connect different ONGC sub-networks together (e.g., connecting a LAN in Delhi HQ to a LAN in the Assam Asset).
- **Switches:** Connect computers, servers, and printers within the same local building.
- **Firewalls:** Protect the corporate intranet from external threats and unauthorized access.
- **VSAT/Microwave Antennas:** Vital for offshore platforms (like Mumbai High) to communicate with onshore hubs. [\[1, 2, 3, 4, 5, 6\]](#)

3. Types of Networks

ONGC utilizes a hybrid of network scopes:

- **LAN (Local Area Network):** Connects computers within a single facility (e.g., Nazira or Sivasagar field offices). [\[1, 2, 3, 4\]](#)
- **WAN (Wide Area Network):** Spans across vast geographical areas to link all regional offices, assets, and the corporate headquarters in Delhi. [\[1, 2, 3, 4, 5\]](#)
- **SCADA (Supervisory Control and Data Acquisition):** A specialized telemetry network used to monitor and control remote drilling operations, pipelines, and production facilities in real-time. [\[1, 2, 3, 4, 5\]](#)

4. Essential Protocols

Standard transmission rules ensure network stability: [\[1\]](#)

- **TCP/IP:** The core communication language of the intranet.
- **VoIP (Voice over IP):** Enables cost-effective, continuous internal voice communication across all remote ONGC assets